A Retrieval-Augmented Generation Chatbot for Accessing Reliable Information on Violence Against Women and Girls in the Philippines

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***Abstract*—Violence against women and girls remains a significant concern, and access to reliable information about reporting mechanisms, protection measures, and support services can be difficult when relevant information is distributed across different laws, government guidelines, and institutional resources. This study proposes a Retrieval-Augmented Generation (RAG)-based chatbot designed to provide accessible and grounded information concerning violence against women and girls in the Philippines, in support of Sustainable Development Goal 5, particularly Target 5.2. The proposed system will use a curated knowledge base consisting of authoritative Philippine documents, including Republic Act No. 9262, the Barangay VAW Desk Handbook, and Philippine Commission on Women resources. The chatbot will retrieve relevant passages from these sources and use a language model to generate understandable responses while limiting generated information to the retrieved context. Meta's Llama 3.1 8B Instant, deployed through the Groq API, is proposed as the initial generation model. The system will be evaluated using a preliminary benchmark question set covering legal provisions, reporting procedures, protection measures, referral processes, and available support services. Evaluation will consider response accuracy, retrieval relevance, groundedness, and completeness, with out-of-scope questions included to assess the chatbot's ability to avoid unsupported claims. The study aims to determine how effectively a RAG-based approach can provide accurate and grounded information from authoritative Philippine sources while maintaining a focused and manageable scope for academic research.**

Keywords—Retrieval-Augmented Generation (RAG), violence against women and girls, SDG 5, Target 5.2, information access, reporting mechanisms, protection services, support services, Philippine government resources, Llama 3.1.

I. Selected UN SDG and Justification

The selected Sustainable Development Goal (SDG) for this study is SDG 5 – Achieve Gender Equality and Empower All Women and Girls, with particular emphasis on Target 5.2, which aims to eliminate all forms of violence against women and girls in both public and private spheres, including trafficking and sexual and other forms of exploitation [1]. The area under consideration within the target is preventing violence against women and girls, especially ensuring access to accurate information on how they can report any form of violence. It is important to note that violence against women continues to be a major challenge throughout the globe. According to UN, nearly one-third of women globally, roughly 840 million women and girls, have suffered either sexual or physical violence from their intimate partner or sexual violence from a stranger in their lifetime [1].

In the Philippines, according to the 2022 Philippine National Demographic and Health Survey, approximately 34% of women who suffered physical or sexual violence sought help to stop the violence, while 42% neither sought any help nor informed anyone about their suffering [3]. The data provided above proves the significance of providing people with necessary information about ways of stopping violence and possible measures for protection and support. Thus, the development of an information chatbot using the Retrieval-Augmented Generation technology (RAG) may help people get access to useful information from reliable sources. According to the suggested idea, an RAG chatbot will retrieve information from the set of selected authoritative laws, guidelines, frequently asked questions, and other official publications of the Philippines before responding to users' questions. Contrary to a regular chatbot which mostly uses its previously trained knowledge, the proposed chatbot will be grounded in the retrieved information; hence, the study will have an opportunity to analyze the correctness and groundedness of the provided information. In view of the above, SDG 5 Target 5.2 was chosen since it directly relates to the suggested information issue.

II. Narrow Problem Statement

Level	Group Output
Broad domain	Violence against women and girls
SDG-related issue	Access to reliable information about preventing, reporting, and responding to violence against women and girls, aligned with **SDG 5 Target 5.2**
Target users	Women and girls who may be experiencing or at risk

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	of experiencing violence, as well as individuals seeking information on behalf of someone experiencing violence
Narrow problem	Users may have difficulty finding reliable and understandable information about where and how to report violence and what protection or support services are available. Information may be distributed across different government websites, laws, guidelines, and service-provider resources, making it difficult to identify the appropriate source or next step.
Proposed chatbot focus	A RAG chatbot that answers questions about reporting procedures, available protection/support services, relevant laws and policies, and appropriate government or institutional resources using a curated collection of authoritative documents.

Victimized women and girls, as well as people seeking assistance on their behalf, may struggle to find accurate and clear information about where and how to report cases, available protection measures and support services, and ways in which they can seek assistance. This information is scattered across various sources within the government, legislation, policy, and service provider documents, thus making it hard for the user to locate the right document and information source. This paper, therefore, suggests the development of a RAG Chatbot that will retrieve information from a set of documents that are curated by experts on violence against women and girls to provide assistance to users with their inquiries about reporting channels, protection, services, and policies..

III. Primary Research Question

"How effectively can a RAG-based chatbot retrieve and provide accurate and grounded answers to questions about reporting mechanisms, protection measures, and support services for violence against women and girls using a curated collection of authoritative Philippine documents? "

The research question, "How effectively can a RAG-based chatbot provide accurate and grounded information about reporting, protection, and support services for violence against women and girls using authoritative Philippine documents?", has been formulated based on the narrow information problem identified in the study. In fact, the question is concerned with the challenge of obtaining accurate and reliable information about reporting mechanisms, protection services, and support services, but not with addressing the problem of violence against women and girls at large. Thus, the research question is relevant to the proposed RAG-based chatbot because it involves the examination of its capacity to access information from the knowledge base and employ it when creating answers. Moreover, the question is measurable since the performance of the chatbot can be evaluated by means of a predefined set of questions and criteria (e.g., answer accuracy, retrieval relevance, groundedness, and completeness). Lastly, the limitation of the knowledge source to the authoritative Philippine documents allows narrowing the scope of the study, which makes it possible to identify expected answers and corresponding passages for comparison.

IV. Initial Knowledge-Base Data Sources

Source Name / Organization	Document Type	Information Description
Republic Act No. 9262 – Anti-Violence Against Women and Their Children Act of 2004 / Philippine Government	Philippine law / legal document	Provides the legal definitions, forms of violence covered, protection measures, and responsibilities of relevant authorities in responding to violence against women and their children.
Barangay VAW Desk Handbook / Philippine Commission on Women (PCW) and partner government agencies	Government handbook / procedural guide	Provides guidance on the roles and responsibilities of Barangay VAW Desk officers, including handling, recording, referral, and assistance for VAW cases.
FAQs on Republic Act No. 9262 / Philippine Commission on Women (PCW)	Official FAQ / webpage	Presents commonly asked questions and answers concerning RA 9262, including information about VAWC,

		protection, and where victims can seek assistance.

The selected three sources were chosen for the following reasons since they offer different types of information which could be helpful for solving the research problem. Firstly, Republic Act No. 9262 is the primary legal source providing definitions, types of acts protected by the law, methods of protection and responsibilities of authorities. Secondly, Barangay VAW Desk Handbook is a source supplementing the law with practical information on the procedures for dealing with cases of violence against women at barangay level. Lastly, RA 9262 FAQ by Philippine Commission on Women provides information in the form of questions and answers. This fact enables the use of this source as a means of presenting potential questions which users could address. As a result, the usage of these sources will help us to receive information from the primary legal documents and guidelines as well as keep the knowledge base small enough. Finally, the selected sources seem realistic for acquisition through official Philippine government information resources.

V. Initial Evaluation Data Plan

To evaluate the effectiveness of the proposed Retrieval-Augmented Generation (RAG) chatbot, a preliminary benchmark test question set will be developed to assess the accuracy, relevance, groundedness, and completeness of its responses based on the retrieved context [7], [8]. Realistic user questions will be formulated based on common information needs concerning violence against women and girls, particularly questions related to the legal provisions of Republic Act No. 9262, Barangay Violence Against Women (VAW) Desk procedures, reporting mechanisms, protection measures, and available support services. The evaluation dataset will include different question types to assess the chatbot across a range of information needs. Factual questions will address definitions, covered acts, and legal provisions under RA 9262 [5]. Procedural questions will focus on the roles and responsibilities of Barangay VAW Desks, including case handling, documentation, and referral procedures [4].

Support- and reporting-related questions will examine the chatbot's ability to identify appropriate agencies, offices, and available assistance based on the information provided by the Philippine Commission on Women (PCW) [5]. The test set may also include questions expressed using common or conversational Filipino and Taglish phrasing to determine whether the chatbot can retrieve the appropriate information despite variations in how users formulate their questions. Ground-truth reference answers and supporting passages will be established by pairing each test question with relevant passages from the selected authoritative documents. Expected answers will be manually formulated based on these verified passages and will serve as references for evaluating the accuracy and groundedness of the chatbot's generated responses.

In addition, several out-of-scope questions will be intentionally included when their answers are not available in the knowledge base. These questions will determine whether the chatbot can recognize insufficient information and provide an appropriate response instead of generating unsupported claims. The preliminary evaluation dataset will be developed and manually reviewed by the researchers in a subsequent stage of the study, and no claim is made that the evaluation data has already been collected.

VI. Possible Language Model and Justification

The proposed language model for the generation component of the Retrieval-Augmented Generation (RAG) pipeline is Meta's Llama 3.1 8B Instant (llama-3.1-8b-instant), deployed through the Groq LPU API platform [5]. Factual adherence is particularly important when providing information concerning violence against women and girls, as inaccurate or unsupported responses about reporting mechanisms, protection measures, legal provisions, or support services may mislead users seeking assistance. Llama 3.1 8B Instant is proposed because its instruction-following capabilities and large context window can support the generation of responses based on the information retrieved from the knowledge base [5]. Its 8-billion-parameter size and 128K-token context window also provide a practical balance between language understanding, multi-document context processing, and response generation for the proposed academic prototype [5].

In the proposed RAG architecture, the model will receive the user's query together with relevant passages retrieved from the selected authoritative Philippine documents, including Republic Act No. 9262, the Barangay VAW Desk Handbook, and Philippine Commission on Women (PCW) resources [1]–[3]. Its expected role is to synthesize the retrieved legal and procedural information into clear and understandable responses for users seeking information about reporting mechanisms, protection measures, and available support services. The generation component will be guided by system instructions requiring responses to remain grounded in the retrieved context and to avoid introducing unsupported information. When the requested information is not available in the knowledge base, the chatbot will be expected to indicate that the available sources do not provide sufficient information rather than generate an unsupported answer.

From a practical perspective, deploying Llama 3.1 8B Instant through the Groq API provides an accessible approach for prototyping the chatbot without requiring the researchers to maintain local GPU hardware. Its use also allows the researchers to focus on retrieval, prompt design, and evaluation while using an externally hosted inference platform [5]. The model will initially be used through prompt-based generation rather than assuming that fine-tuning is necessary. Its ability to process varied natural-language inputs may also support questions expressed in English, Filipino, or Taglish, although this capability will be evaluated rather than assumed. The selected model therefore serves as a preliminary language-model choice for the proposed RAG prototype,

and its response quality will be empirically evaluated using the study's benchmark questions and compared with alternative models where feasible.

VII. Group Reflection

This problem is narrow enough since the bot will deal specifically with the issue of providing information on the topic of violence against women and girls including reporting, protecting and support services in the Philippines. Instead of trying to cover all possible problems of gender equality and violence against women, the group will limit itself only to the relevant documents from the Philippine government sources like the Republic Act No. 9262, Barangay VAW Desk Handbook and resources of the Philippine Commission on Women. The bot will be tested and evaluated by means of limited number of test questions on the topics of legal issues, reporting, protection, referrals, and support services. It is possible to limit the knowledge base and create an access to the required information by using RAG technology in order to provide answers based on the selected documents.

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